Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Peter DeVisser Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
21268 E Avena RD	Escalon	San Joaquin	95320

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1950

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X205-X230-XX01-XXXX

B. OPERATORS

DeVisser, Peter

Operator name: DeVisser, Peter

Telephone no.: (209) 838-7383 Landline, (209) 581-1692 Cellular

Mailing Address Number and Street	City	State	Zip Code
21268 E Avena RD	Escalon	CA	95320

This operator is responsible for paying permit fees.

C. OWNERS

DeVisser, Peter

Legal owner name: DeVisser, Peter

Telephone no.: (209) 838-7383 Landline, (209) 581-1692 Cellular

Mailing Address Number and Street	City	State	Zip Code
21268 E Avena RD	Escalon	CA	95320

This owner is responsible for paying permit fees.

DeVisser Trust, Peter J. and Rena

Legal owner name:	DeVisser Trust, Peter J. and Rena	Telephone no.:	(209) 838-7383	(209) 581-1692
			Landline	Cellular
21268 E Avena RD	Escalon		CA	95320
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	20	250	50	0
Number under roof	680	91	0	0	0	0
Maximum number	680	91	20	250	50	0
Average number	680	91	20	250	50	0
Avg live weight (lbs)	1,400	1,450	1,000	700		

Predominant milk cow breed: Holstein

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 21,923.50 *tons per reporting period*

Total nitrogen from manure: 278,061.13 *lbs per reporting period*

After ammonia losses (30% loss applied): 194,642.79 *lbs per reporting period*

Total phosphorus from manure: 46,277.29 *lbs per reporting period*

Total potassium from manure: 132,460.07 *lbs per reporting period*

Total salt from manure: 341,103.45 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 15,618,000 *gallons*

Total nitrogen generated: 82,786.24 *lbs*

Total phosphorus generated: 10,800.23 *lbs*

Total potassium generated: 121,293.27 *lbs*

Total salt generated: 703,171.06 *lbs*

	15,618,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	15,618,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
IW 1	Ground water
IW 2	Ground water
IW 3	Ground water
IW Almonds	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
05/09/2023	Solid commercial fertilizer *UN 32*	10.12 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000
06/23/2023	Solid commercial fertilizer *4-0-0*	26.87 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
11/01/2023	Solid commercial fertilizer *Can 17*	20.54 *ton*	Dry-weight	0.1	17.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	15,594.39	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	15,594.39	0.00	0.00	0.00

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	20	20	2	both	X205-X230-XX01-XXXX
Field 2	95	95	2	process wastewater	X205-X230-XX01-XXXX
Field 3	60	60	2	process wastewater	X205-X220-XX10-XXXX
Field 4	25	25	2	both	X205-X230-XX01-XXXX
Field 5	15	15	2	manure	X205-X230-XX01-XXXX
Field P1	40	40	1	none	X205-X220-XX10-XXXX
Field P2	32	32	1	none	X205-X220-XX09-XXXX
Field P3	20	20	1	none	X205-X220-XX10-XXXX
Field P4	10	10	1	manure	X205-X230-XX01-XXXX
Totals for areas that were used for application	225	225	11		
Totals for areas that were not used for application	92	92	3		
Land application area totals	317	317	14		

B. CROPS AND HARVESTS

Field 1

Field name: Field 1

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 20 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	240.00 *ton*	As-is		56.4	8,740.00	1,300.00	13,200.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.00	209.76	31.20	316.80	1,067.33

Field 1

06/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 20 Plant date: 06/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	622.00 *ton*	As-is		72.2	3,670.00	430.00	4,800.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	31.10	228.27	26.75	298.56	1,296.87

Field 2

Field name: Field 2

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 95 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	1,504.80 *ton*	As-is		72.0	5,600.00	1,030.00	9,940.00		13.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.84	177.41	32.63	314.90	1,170.89

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 95 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	2,774.00 *ton*	As-is		67.6	3,790.00	430.00	4,210.00		6.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.20	221.34	25.11	245.86	1,192.06

Field 3

Field name: Field 3

05/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 05/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	780.00 *ton*	As-is		58.4	7,000.00	1,290.00	12,300.00		10.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	13.00	182.00	33.54	319.80	1,081.60

07/01/2023: Corn, silage

Crop: Corn, silage Acres planted: 60 Plant date: 07/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	1,788.00 *ton*	As-is		68.6	4,110.00	470.00	5,220.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	29.80	244.96	28.01	311.11	1,347.44

Field 4

Field name: Field 4

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 25 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	390.00 *ton*	As-is		65.0	5,070.00	1,090.00	11,900.00		11.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.60	158.18	34.01	371.28	1,244.88

Field 4

06/24/2023: Corn, silage

Crop: Corn, silage Acres planted: 25 Plant date: 06/24/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	762.50 *ton*	As-is		69.8	3,710.00	360.00	4,840.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.50	226.31	21.96	295.24	1,252.70

Field 5

Field name: Field 5

10/31/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/31/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/17/2023	243.00 *ton*	As-is		73.6	6,080.00	1,090.00	13,000.00		12.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.20	196.99	35.32	421.20	1,034.99

06/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	457.50 *ton*	As-is		69.8	3,710.00	360.00	4,840.00		6.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	30.50	226.31	21.96	295.24	1,252.70

Field P1

Field name: Field P1

02/10/2020: Almond Field Run

Crop: Almond Field Run Acres planted: 40 Plant date: 02/10/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/08/2023	126.00 *ton*	As-is		9.2	23,800.00	2,130.00	25,500.00		16.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	129.00	21.90	139.50	0.00
Total actual harvest content	3.15	149.94	13.42	160.65	961.03

Field P2

Field name: Field P2

02/10/2020: Almond Field Run

Crop: Almond Field Run Acres planted: 32 Plant date: 02/10/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/10/2023	93.80 *ton*	As-is		9.2	23,800.00	2,130.00	25,500.00		16.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	3.00	129.00	21.90	139.50	0.00
Total actual harvest content	2.93	139.53	12.49	149.49	894.29

Field P3

Field name: Field P3

Field P3

12/30/2018: Almond Field Run

Crop: Almond Field Run Acres planted: 20 Plant date: 12/30/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/07/2023	97.40 *ton*	As-is		10.4	15,100.00	2,000.00	21,600.00		11.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	4.87	147.07	19.48	210.38	986.16

Field P4

Field name: Field P4

01/01/2023: Pasture

Crop: Pasture Acres planted: 10 Plant date: 01/01/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/24/2023	150.00 *ton*	As-is		72.7	11,100.00	1,260.00	21,600.00		13.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	15.00	276.00	39.00	283.50	0.00
Total actual harvest content	15.00	333.00	37.80	648.00	1,130.22

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 1 - 10/29/2022: Oats, silage-soft dough

Field name: Field 1

Crop: Oats, silage-soft dough — Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	256.96	125.84	454.08	2,208.80	200.00 *ton*
Application event totals		256.96	125.84	454.08	2,208.80	

Field 1 - 06/23/2023: Corn, silage

Field name: Field 1

Crop: Corn, silage — Plant date: 06/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.94	0.00	0.00	186.26	2,790,000.00 *gal*
Application event totals		1.94	0.00	0.00	186.26	

06/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.38	0.00	0.00	132.18	1,980,000.00 *gal*
Application event totals		1.38	0.00	0.00	132.18	

Field 1 - 06/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	235.18	23.98	367.42	2,205.28	924,000.00 *gal*
Application event totals		235.18	23.98	367.42	2,205.28	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.76	0.00	0.00	168.24	2,520,000.00 *gal*
Application event totals		1.76	0.00	0.00	168.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.76	0.00	0.00	168.24	2,520,000.00 *gal*
Application event totals		1.76	0.00	0.00	168.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.76	0.00	0.00	168.24	2,520,000.00 *gal*
Application event totals		1.76	0.00	0.00	168.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.69	0.00	0.00	162.23	2,430,000.00 *gal*
Application event totals		1.69	0.00	0.00	162.23	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.63	0.00	0.00	156.22	2,340,000.00 *gal*
Application event totals		1.63	0.00	0.00	156.22	

Field 2 - 10/30/2022: Oats, silage-soft dough

Field 2 - 10/30/2022: Oats, silage-soft dough

Field name: Field 2

Crop: Oats, silage-soft dough

Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	41.81	9.43	35.22	168.57	950,000.00 *gal*
Application event totals		41.81	9.43	35.22	168.57	

02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	189.66	27.36	281.57	1,593.81	3,024,000.00 *gal*
Application event totals		189.66	27.36	281.57	1,593.81	

Field 2 - 06/24/2023: Corn, silage

Field name: Field 2

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.90	0.00	0.00	182.15	12,960,000.00 *gal*
Application event totals		1.90	0.00	0.00	182.15	

Field 2 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	236.30	24.10	369.18	2,215.83	4,410,000.00 *gal*
IW 2	Ground water	1.39	0.00	0.00	132.82	9,450,000.00 *gal*
Application event totals		237.69	24.10	369.18	2,348.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.85	0.00	0.00	177.09	12,600,000.00 *gal*
Application event totals		1.85	0.00	0.00	177.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.78	0.00	0.00	170.77	12,150,000.00 *gal*
Application event totals		1.78	0.00	0.00	170.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.74	0.00	0.00	166.97	11,880,000.00 *gal*
Application event totals		1.74	0.00	0.00	166.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.72	0.00	0.00	164.44	11,700,000.00 *gal*
Application event totals		1.72	0.00	0.00	164.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.65	0.00	0.00	158.12	11,250,000.00 *gal*
Application event totals		1.65	0.00	0.00	158.12	

Field 3 - 05/17/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough

Plant date: 05/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	41.81	9.43	35.22	168.57	600,000.00 *gal*
Application event totals		41.81	9.43	35.22	168.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	191.86	27.68	284.83	1,612.25	1,932,000.00 *gal*
Application event totals		191.86	27.68	284.83	1,612.25	

Field 3 - 07/01/2023: Corn, silage

Field name: Field 3

Crop: Corn, silage

Plant date: 07/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.90	0.00	0.00	182.25	8,190,000.00 *gal*
Application event totals		1.90	0.00	0.00	182.25	

Field 3 - 07/01/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.84	0.00	0.00	176.25	7,920,000.00 *gal*
Application event totals		1.84	0.00	0.00	176.25	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	238.74	24.34	372.99	2,238.70	2,814,000.00 *gal*
IW 2	Ground water	1.40	0.00	0.00	134.19	6,030,000.00 *gal*
Application event totals		240.14	24.34	372.99	2,372.88	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.80	0.00	0.00	172.24	7,740,000.00 *gal*
Application event totals		1.80	0.00	0.00	172.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.78	0.00	0.00	170.24	7,650,000.00 *gal*
Application event totals		1.78	0.00	0.00	170.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.76	0.00	0.00	168.24	7,560,000.00 *gal*
Application event totals		1.76	0.00	0.00	168.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.67	0.00	0.00	160.22	7,200,000.00 *gal*
Application event totals		1.67	0.00	0.00	160.22	

Field 4 - 10/31/2022: Oats, silage-soft dough

Field name: Field 4

Crop: Oats, silage-soft dough

Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater sludge	41.81	9.43	35.22	168.57	250,000.00 *gal*
Application event totals		41.81	9.43	35.22	168.57	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	170.17	24.55	252.63	1,430.00	714,000.00 *gal*
Application event totals		170.17	24.55	252.63	1,430.00	

Field 4 - 06/24/2023: Corn, silage

Field name: Field 4

Crop: Corn, silage

Plant date: 06/24/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	239.57	126.38	350.67	1,881.82	140.00 *ton*
Application event totals		239.57	126.38	350.67	1,881.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Field 4 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	9.13	0.00	0.00	313.64	3,480,000.00 *gal*
Application event totals		9.13	0.00	0.00	313.64	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.66	0.00	0.00	297.42	3,300,000.00 *gal*
Application event totals		8.66	0.00	0.00	297.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.50	0.00	0.00	292.01	3,240,000.00 *gal*
Application event totals		8.50	0.00	0.00	292.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.50	0.00	0.00	292.01	3,240,000.00 *gal*
Application event totals		8.50	0.00	0.00	292.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.50	0.00	0.00	292.01	3,240,000.00 *gal*
Application event totals		8.50	0.00	0.00	292.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.34	0.00	0.00	286.60	3,180,000.00 *gal*
Application event totals		8.34	0.00	0.00	286.60	

Field 4 - 06/24/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	7.71	0.00	0.00	264.97	2,940,000.00 *gal*
Application event totals		7.71	0.00	0.00	264.97	

Field 5 - 10/31/2022: Oats, silage-soft dough

Field name: Field 5

Crop: Oats, silage-soft dough Plant date: 10/31/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	256.96	125.84	454.08	2,208.80	150.00 *ton*
Application event totals		256.96	125.84	454.08	2,208.80	

Field 5 - 06/27/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage Plant date: 06/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	228.16	120.36	333.97	1,792.21	80.00 *ton*
Application event totals		228.16	120.36	333.97	1,792.21	

Field 5 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	9.18	0.00	0.00	315.44	2,100,000.00 *gal*
Application event totals		9.18	0.00	0.00	315.44	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	10.00	0.00	0.00	0.00	
Application event totals		10.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.66	0.00	0.00	297.42	1,980,000.00 *gal*
Application event totals		8.66	0.00	0.00	297.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.40	0.00	0.00	288.40	1,920,000.00 *gal*
Application event totals		8.40	0.00	0.00	288.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.40	0.00	0.00	288.40	1,920,000.00 *gal*
Application event totals		8.40	0.00	0.00	288.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.40	0.00	0.00	288.40	1,920,000.00 *gal*
Application event totals		8.40	0.00	0.00	288.40	

Field 5 - 06/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	8.13	0.00	0.00	279.39	1,860,000.00 *gal*
Application event totals		8.13	0.00	0.00	279.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	7.87	0.00	0.00	270.38	1,800,000.00 *gal*
Application event totals		7.87	0.00	0.00	270.38	

Field P1 - 02/10/2020: Almond Field Run

Field name: Field P1

Crop: Almond Field Run

Plant date: 02/10/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/01/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	34.00	0.00	0.00	0.00	
Application event totals		34.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	37.40	0.00	0.00	0.00	
Application event totals		37.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	4,030,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Field P1 - 02/10/2020: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	4,030,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

05/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	4,030,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	4,030,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

06/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Field P1 - 02/10/2020: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Field P1 - 02/10/2020: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	4,086,000.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Field P2 - 02/10/2020: Almond Field Run

Field name: Field P2

Crop: Almond Field Run

Plant date: 02/10/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
01/02/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	34.00	0.00	0.00	0.00	
Application event totals		34.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	37.40	0.00	0.00	0.00	
Application event totals		37.40	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	3,224,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	3,224,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Field P2 - 02/10/2020: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

05/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	3,224,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	3,224,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

06/09/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

06/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

07/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Field P2 - 02/10/2020: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.14	0.00	0.00	119.34	3,268,800.00 *gal*
Application event totals		1.14	0.00	0.00	119.34	

Field P3 - 12/30/2018: Almond Field Run

Field P3 - 12/30/2018: Almond Field Run

Field name: Field P3

Crop: Almond Field Run

Plant date: 12/30/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.74	0.00	0.00	166.23	2,043,000.00 *gal*
Application event totals		1.74	0.00	0.00	166.23	

10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	1.74	0.00	0.00	166.23	2,043,000.00 *gal*
Application event totals		1.74	0.00	0.00	166.23	

11/03/2022	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	37.40	0.00	0.00	0.00	
Application event totals		37.40	0.00	0.00	0.00	

03/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	42.50	0.00	0.00	0.00	
Application event totals		42.50	0.00	0.00	0.00	

04/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.11	0.00	0.00	116.25	1,990,000.00 *gal*
Application event totals		1.11	0.00	0.00	116.25	

05/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.11	0.00	0.00	116.25	1,990,000.00 *gal*
Application event totals		1.11	0.00	0.00	116.25	

Field P3 - 12/30/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/08/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

05/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.11	0.00	0.00	116.25	1,990,000.00 *gal*
Application event totals		1.11	0.00	0.00	116.25	

06/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

06/06/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

06/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Field P3 - 12/30/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Almonds	Ground water	1.13	0.00	0.00	117.71	2,015,000.00 *gal*
Application event totals		1.13	0.00	0.00	117.71	

Field P4 - 01/01/2023: Pasture

Field P4 - 01/01/2023: Pasture

Field name: Field P4

Crop: Pasture

Plant date: 01/01/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.49	0.00	0.00	223.06	990,000.00 *gal*
Application event totals		6.49	0.00	0.00	223.06	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.69	0.00	0.00	229.82	1,020,000.00 *gal*
Application event totals		6.69	0.00	0.00	229.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	342.24	180.54	500.96	2,688.32	80.00 *ton*
Application event totals		342.24	180.54	500.96	2,688.32	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.69	0.00	0.00	229.82	1,020,000.00 *gal*
Application event totals		6.69	0.00	0.00	229.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.89	0.00	0.00	236.58	1,050,000.00 *gal*
Application event totals		6.89	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.89	0.00	0.00	236.58	1,050,000.00 *gal*
Application event totals		6.89	0.00	0.00	236.58	

Field P4 - 01/01/2023: Pasture

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.89	0.00	0.00	236.58	1,050,000.00 *gal*
Application event totals		6.89	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.89	0.00	0.00	236.58	1,050,000.00 *gal*
Application event totals		6.89	0.00	0.00	236.58	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.69	0.00	0.00	229.82	1,020,000.00 *gal*
Application event totals		6.69	0.00	0.00	229.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.69	0.00	0.00	229.82	1,020,000.00 *gal*
Application event totals		6.69	0.00	0.00	229.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.30	0.00	0.00	216.30	960,000.00 *gal*
Application event totals		6.30	0.00	0.00	216.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.30	0.00	0.00	216.30	960,000.00 *gal*
Application event totals		6.30	0.00	0.00	216.30	

Field P4 - 01/01/2023: Pasture

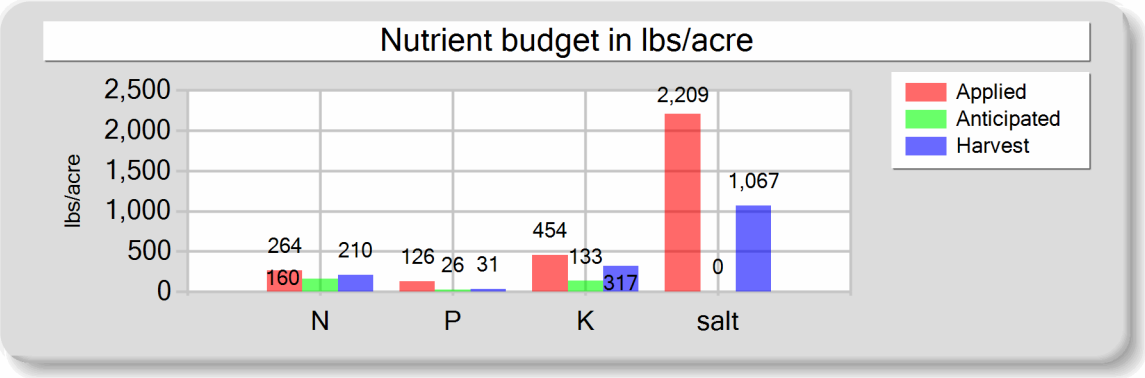
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 1	Ground water	6.30	0.00	0.00	216.30	960,000.00 *gal*
Application event totals		6.30	0.00	0.00	216.30	

B. NUTRIENT BUDGET

Field 1 - 10/29/2022: Oats, silage-soft dough

Field name: Field 1 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	256.96	125.84	454.08	2,208.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.96	125.84	454.08	2,208.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	209.76	31.20	316.80	1,067.33
Nutrient balance	54.20	94.64	137.28	1,141.47
Applied to removed ratio	1.26	4.03	1.43	2.07

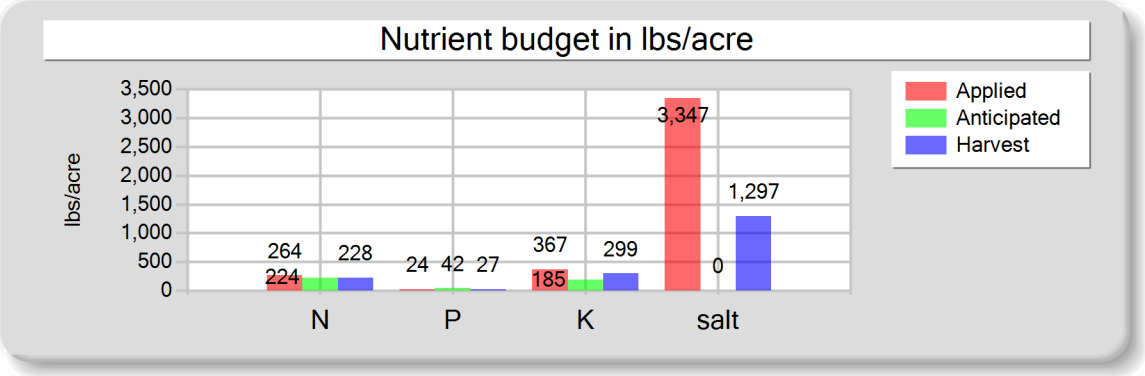
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 1 - 06/23/2023: Corn, silage

Field name: Field 1 Crop: Corn, silage Plant date: 06/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	235.18	23.98	367.42	2,205.28
Fresh water	11.92	0.00	0.00	1,141.60
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	264.09	23.98	367.42	3,346.88
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	228.27	26.75	298.56	1,296.87
Nutrient balance	35.82	-2.77	68.86	2,050.01
Applied to removed ratio	1.16	0.90	1.23	2.58

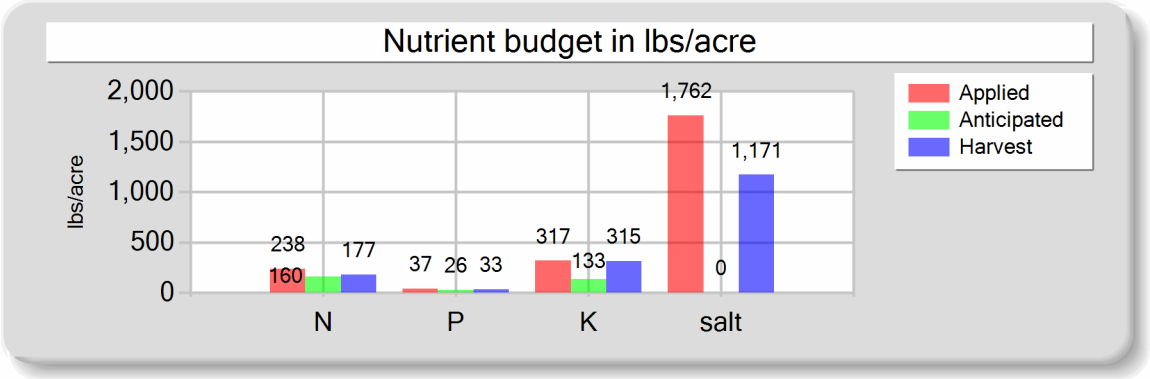
Fresh water applied	
17,100,000.00	*gallons*
629.73	*acre-inches*
31.49	*inches/acre*

Process wastewater applied	
924,000.00	*gallons*
34.03	*acre-inches*
1.70	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 10/30/2022: Oats, silage-soft dough

Field name: Field 2 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	231.47	36.79	316.79	1,762.38
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	238.47	36.79	316.79	1,762.38
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	177.41	32.63	314.90	1,170.89
Nutrient balance	61.06	4.16	1.89	591.48
Applied to removed ratio	1.34	1.13	1.01	1.51

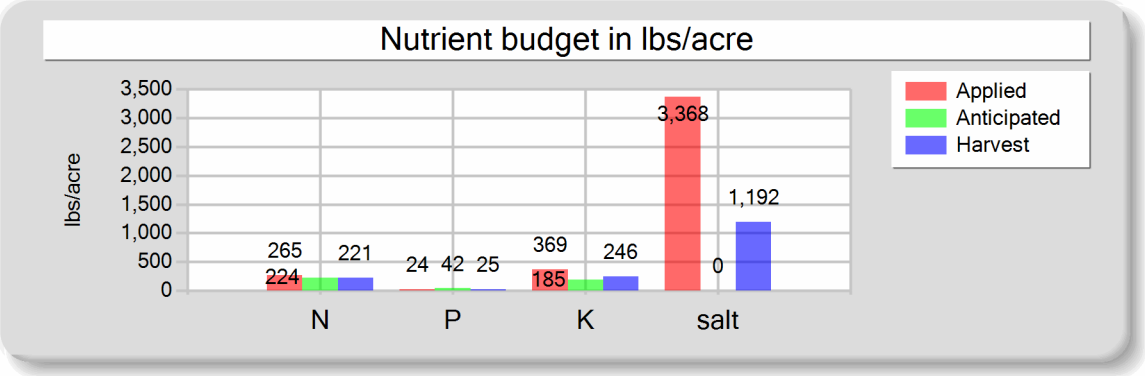
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
3,974,000.00	*gallons*
146.35	*acre-inches*
1.54	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 2 - 06/24/2023: Corn, silage

Field name: Field 2 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	236.30	24.10	369.18	2,215.83
Fresh water	12.03	0.00	0.00	1,152.35
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	265.33	24.10	369.18	3,368.18
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	221.34	25.11	245.86	1,192.06
Nutrient balance	44.00	-1.02	123.31	2,176.12
Applied to removed ratio	1.20	0.96	1.50	2.83

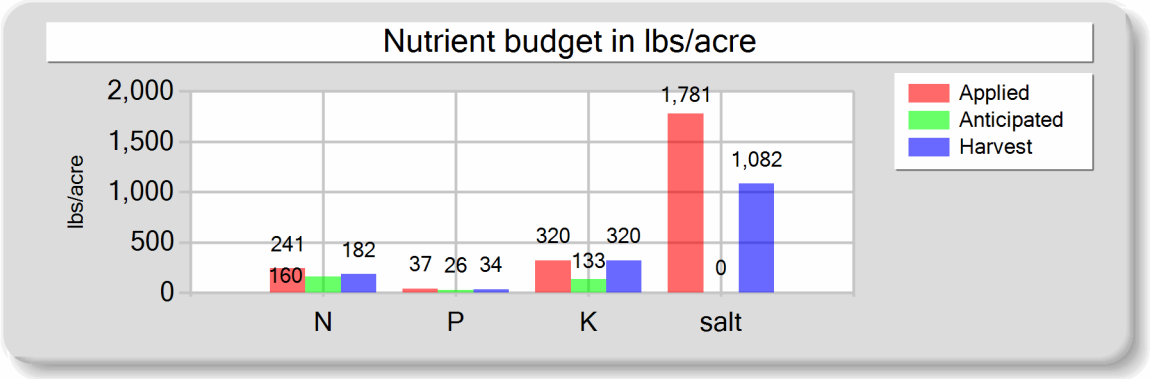
Fresh water applied	
81,990,000.00	*gallons*
3,019.41	*acre-inches*
31.78	*inches/acre*

Process wastewater applied	
4,410,000.00	*gallons*
162.41	*acre-inches*
1.71	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 05/17/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 05/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	233.67	37.11	320.05	1,780.82
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.67	37.11	320.05	1,780.82
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	182.00	33.54	319.80	1,081.60
Nutrient balance	58.67	3.57	0.25	699.22
Applied to removed ratio	1.32	1.11	1.00	1.65

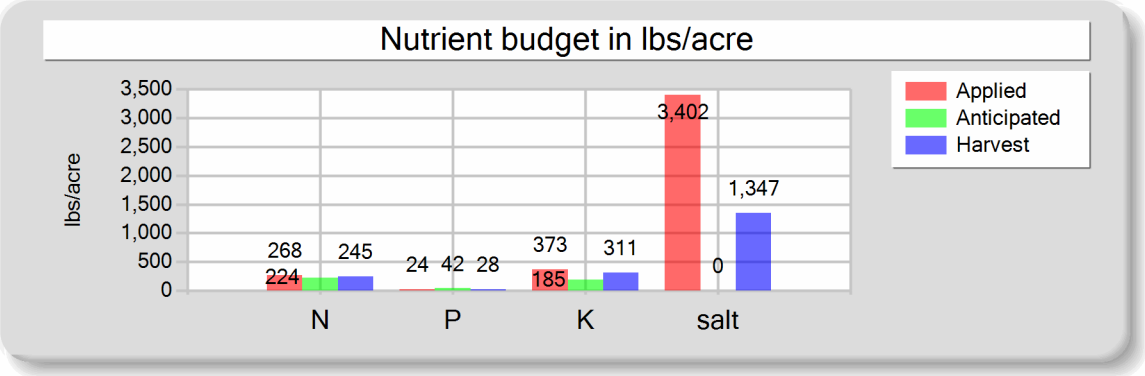
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
2,532,000.00	*gallons*
93.24	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 3 - 07/01/2023: Corn, silage

Field name: Field 3 Crop: Corn, silage Plant date: 07/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.74	24.34	372.99	2,238.70
Fresh water	12.15	0.00	0.00	1,163.63
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	267.89	24.34	372.99	3,402.32
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	244.96	28.01	311.11	1,347.44
Nutrient balance	22.93	-3.67	61.87	2,054.89
Applied to removed ratio	1.09	0.87	1.20	2.53

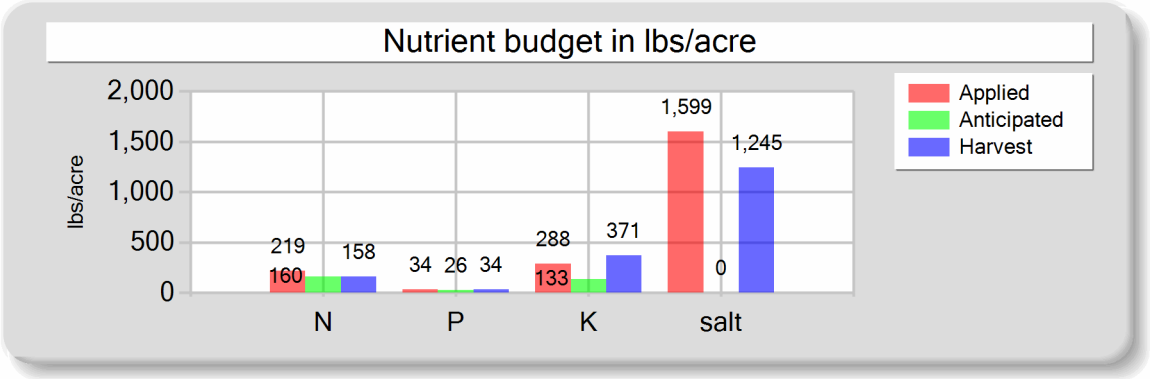
Fresh water applied	
52,290,000.00	*gallons*
1,925.66	*acre-inches*
32.09	*inches/acre*

Process wastewater applied	
2,814,000.00	*gallons*
103.63	*acre-inches*
1.73	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 10/31/2022: Oats, silage-soft dough

Field name: Field 4 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	211.98	33.98	287.85	1,598.57
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	218.98	33.98	287.85	1,598.57
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	158.18	34.01	371.28	1,244.88
Nutrient balance	60.79	-0.03	-83.43	353.69
Applied to removed ratio	1.38	1.00	0.78	1.28

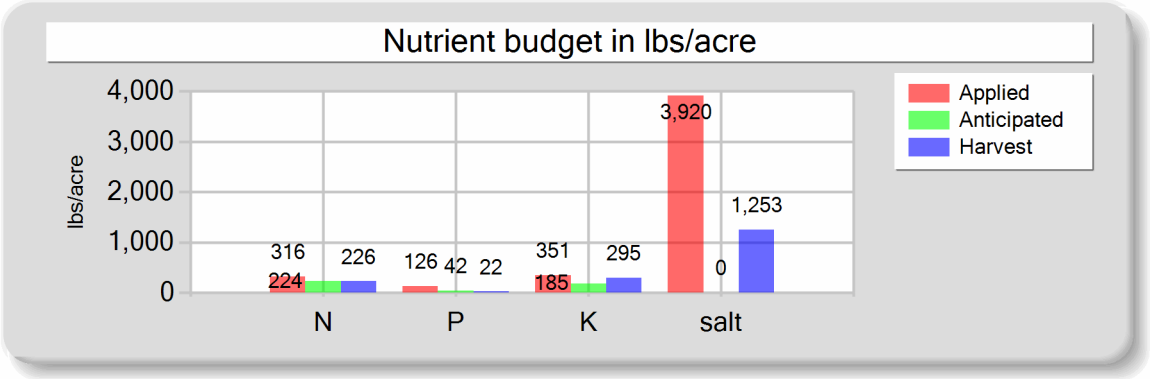
Fresh water applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Process wastewater applied	
964,000.00	*gallons*
35.50	*acre-inches*
1.42	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 4 - 06/24/2023: Corn, silage

Field name: Field 4 Crop: Corn, silage Plant date: 06/24/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	239.57	126.38	350.67	1,881.82
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	59.35	0.00	0.00	2,038.65
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	315.92	126.38	350.67	3,920.47
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	226.31	21.96	295.24	1,252.70
Nutrient balance	89.61	104.42	55.43	2,667.78
Applied to removed ratio	1.40	5.76	1.19	3.13

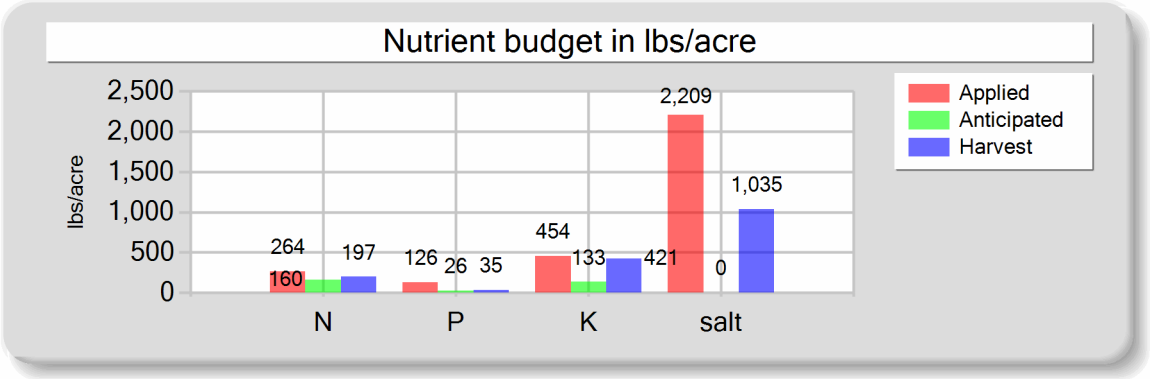
Fresh water applied	
22,620,000.00	*gallons*
833.02	*acre-inches*
33.32	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field 5 - 10/31/2022: Oats, silage-soft dough

Field name: Field 5 Crop: Oats, silage-soft dough Plant date: 10/31/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	256.96	125.84	454.08	2,208.80
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	263.96	125.84	454.08	2,208.80
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	196.99	35.32	421.20	1,034.99
Nutrient balance	66.97	90.52	32.88	1,173.81
Applied to removed ratio	1.34	3.56	1.08	2.13

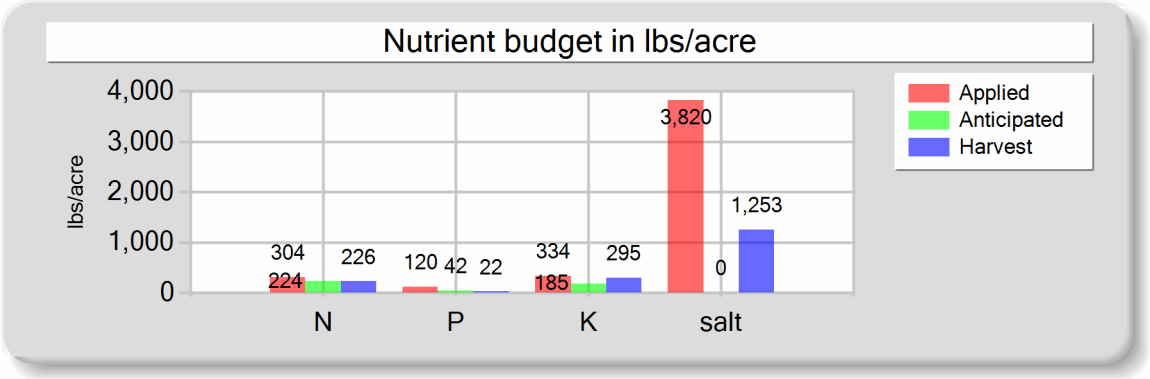
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

Field 5 - 06/27/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 06/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	10.00	0.00	0.00	0.00
Dry manure	228.16	120.36	333.97	1,792.21
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	59.03	0.00	0.00	2,027.84
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.19	120.36	333.97	3,820.05
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	226.31	21.96	295.24	1,252.70
Nutrient balance	77.88	98.40	38.73	2,567.35
Applied to removed ratio	1.34	5.48	1.13	3.05

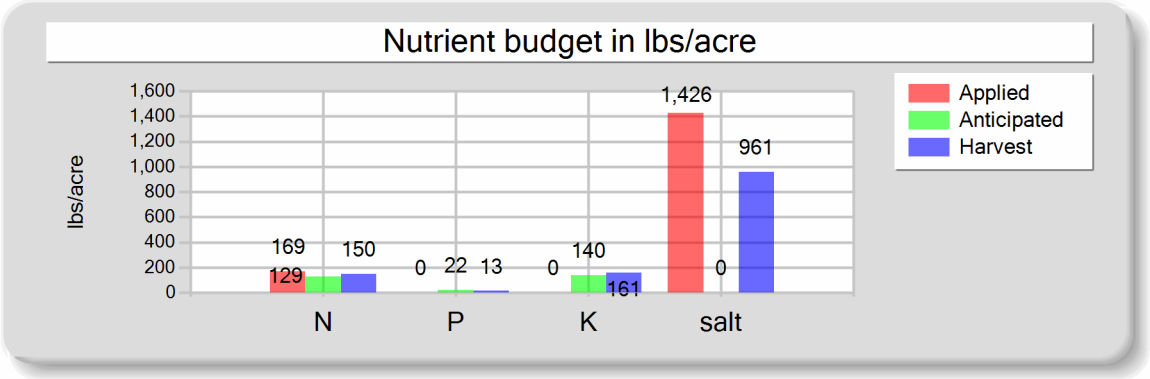
Fresh water applied	
13,500,000.00	*gallons*
497.16	*acre-inches*
33.14	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P1 - 02/10/2020: Almond Field Run

Field name: Field P1 Crop: Almond Field Run Plant date: 02/10/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	141.80	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.64	0.00	0.00	1,425.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	169.44	0.00	0.00	1,425.56
Anticipated crop nutrient removal	129.00	21.90	139.50	0.00
Actual crop nutrient removal	149.94	13.42	160.65	961.03
Nutrient balance	19.50	-13.42	-160.65	464.53
Applied to removed ratio	1.13	0.00	0.00	1.48

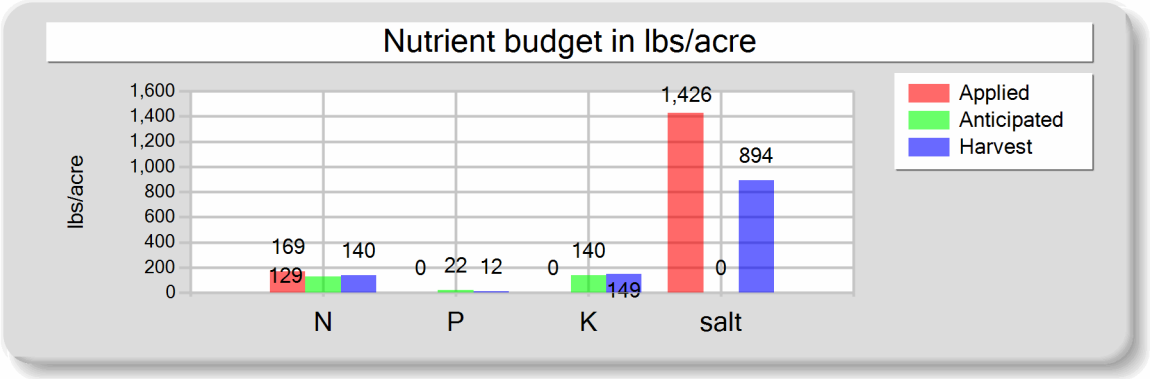
Fresh water applied	
48,808,000.00	*gallons*
1,797.43	*acre-inches*
44.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P2 - 02/10/2020: Almond Field Run

Field name: Field P2 Crop: Almond Field Run Plant date: 02/10/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	141.80	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.64	0.00	0.00	1,425.56
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	169.44	0.00	0.00	1,425.56
Anticipated crop nutrient removal	129.00	21.90	139.50	0.00
Actual crop nutrient removal	139.53	12.49	149.49	894.29
Nutrient balance	29.92	-12.49	-149.49	531.27
Applied to removed ratio	1.21	0.00	0.00	1.59

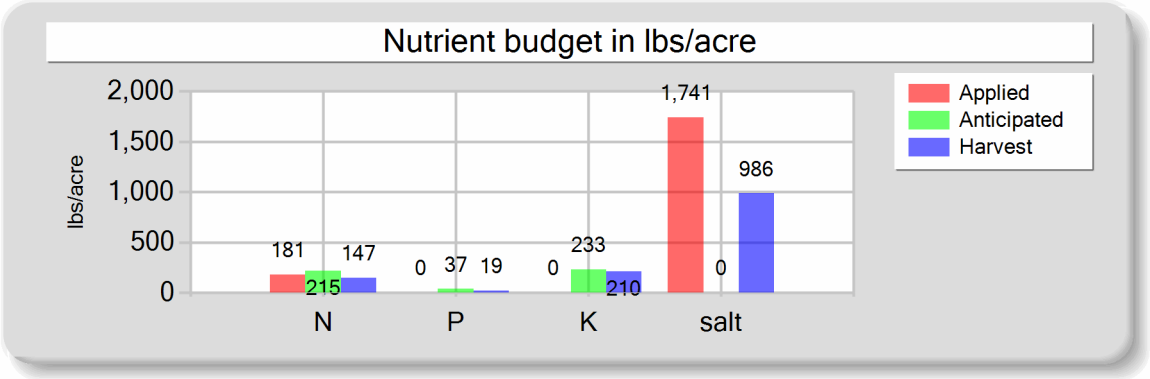
Fresh water applied	
39,046,400.00	*gallons*
1,437.95	*acre-inches*
44.94	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P3 - 12/30/2018: Almond Field Run

Field name: Field P3 Crop: Almond Field Run Plant date: 12/30/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	150.30	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	16.96	0.00	0.00	1,740.55
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	181.26	0.00	0.00	1,740.55
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	147.07	19.48	210.38	986.16
Nutrient balance	34.18	-19.48	-210.38	754.39
Applied to removed ratio	1.23	0.00	0.00	1.76

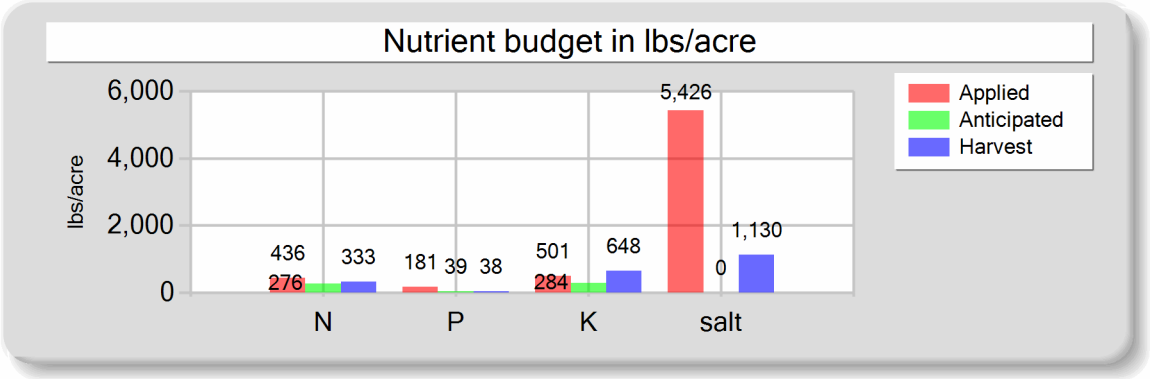
Fresh water applied	
28,191,000.00	*gallons*
1,038.18	*acre-inches*
51.91	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Field P4 - 01/01/2023: Pasture

Field name: Field P4 Crop: Pasture Plant date: 01/01/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	342.24	180.54	500.96	2,688.32
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	79.69	0.00	0.00	2,737.58
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	435.93	180.54	500.96	5,425.90
Anticipated crop nutrient removal	276.00	39.00	283.50	0.00
Actual crop nutrient removal	333.00	37.80	648.00	1,130.22
Nutrient balance	102.93	142.74	-147.04	4,295.68
Applied to removed ratio	1.31	4.78	0.77	4.80

Fresh water applied	
12,150,000.00	*gallons*
447.44	*acre-inches*
44.74	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 38.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	34,500.00	18,200.00	50,500.00							27.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,200.00	14,300.00	51,600.00							25.10
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 56.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,400.00	11,400.00	33,500.00							23.40
DL	100.00	100.00	100.00							0.01

Settling Basin

Sample and source description: Settling Basin

Sample date: 10/09/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,200.00	7,280.00	21,500.00							42.50
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

Slurry

Sample and source description: Slurry

Sample date: 09/30/2022 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	501.00	71.40	0.00	0.00	113.00	422.00	121.00	79.70	151.00	5,180.00	0.00	298.00	243.00	40,600.00	2,020
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	50.00	1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	714.00	398.00	0.00	0.00	103.00	1,060.00								7,680.00	6,000
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	652.00	375.00	0.00	0.00	16.50	128.00								7,420.00	5,020
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	610.00	414.00	0.00	0.00	62.20	953.00								7,740.00	5,720
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

Slurry

Sample and source description: Slurry

Sample date: 10/24/2023 Material type: Process wastewater sludge Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	762.00	207.00	0.00	0.00	90.20	1,010.00								6,660.00	2,520
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	683.00	487.00	0.00	0.00	60.40	1,070.00								7,930.00	7,480
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

IW 1

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.86	0.00	7.86								422.00	270
DL	0.50	0.50	0.50								1.00	10

IW 2

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.67	0.00	1.67								212.00	160
DL	0.50	0.50	0.50								1.00	10

IW 3

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	4.90	0.00	4.70								398.00	263
DL	0.50	0.50	0.50								1.00	10

IW Almonds

Fresh Water

Sample description: Fresh Water

Sample date: 10/25/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.34	0.00	1.30								196.00	140
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

Field 1 - 10/29/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,740.00	1,300.00	13,200.00		10.20
DL	100.00	100.00	100.00		0.01

Field 1 - 06/23/2023: Corn, silage

Field 1 - 06/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,670.00	430.00	4,800.00		7.50
DL	100.00	100.00	100.00		0.01

Field 2 - 10/30/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,600.00	1,030.00	9,940.00		13.20
DL	100.00	100.00	100.00		0.01

Field 2 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,790.00	430.00	4,210.00		6.30
DL	100.00	100.00	100.00		0.01

Field 3 - 05/17/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 58.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,000.00	1,290.00	12,300.00		10.00
DL	100.00	100.00	100.00		0.01

Field 3 - 07/01/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,110.00	470.00	5,220.00		7.20
DL	100.00	100.00	100.00		0.01

Field 4 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,070.00	1,090.00	11,900.00		11.40
DL	100.00	100.00	100.00		0.01

Field 4 - 06/24/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,710.00	360.00	4,840.00		6.80
DL	100.00	100.00	100.00		0.01

Field 5 - 10/31/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/22/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 73.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,080.00	1,090.00	13,000.00		12.10
DL	100.00	100.00	100.00		0.01

Field 5 - 06/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,710.00	360.00	4,840.00		6.80
DL	100.00	100.00	100.00		0.01

Field P1 - 02/10/2020: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	2,130.00	25,500.00		16.80
DL	100.00	100.00	100.00		0.01

Field P2 - 02/10/2020: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 9.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,800.00	2,130.00	25,500.00		16.80
DL	100.00	100.00	100.00		0.01

Field P3 - 12/30/2018: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 10.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,100.00	2,000.00	21,600.00		11.30
DL	100.00	100.00	100.00		0.01

Field P4 - 01/01/2023: Pasture

Pasture

Sample and source description: Pasture

Sample date: 04/24/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 72.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,100.00	1,260.00	21,600.00		13.80
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

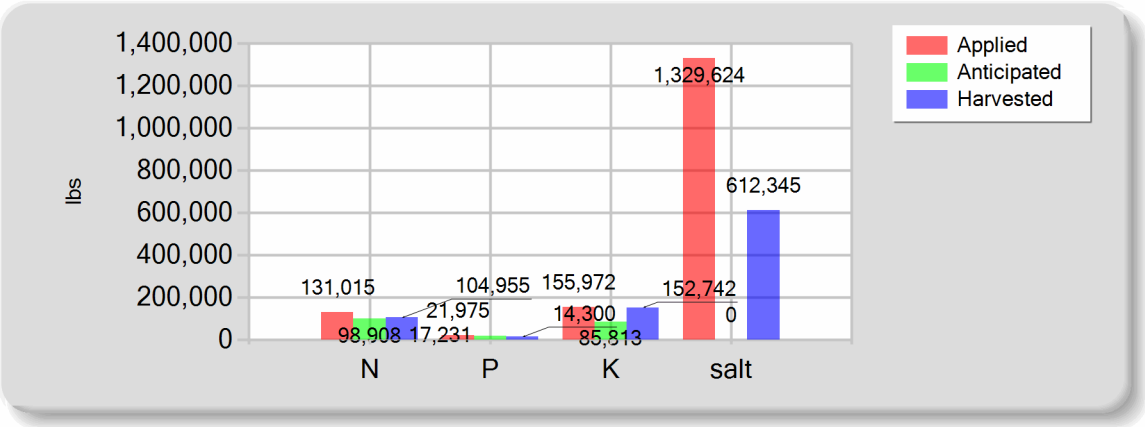
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

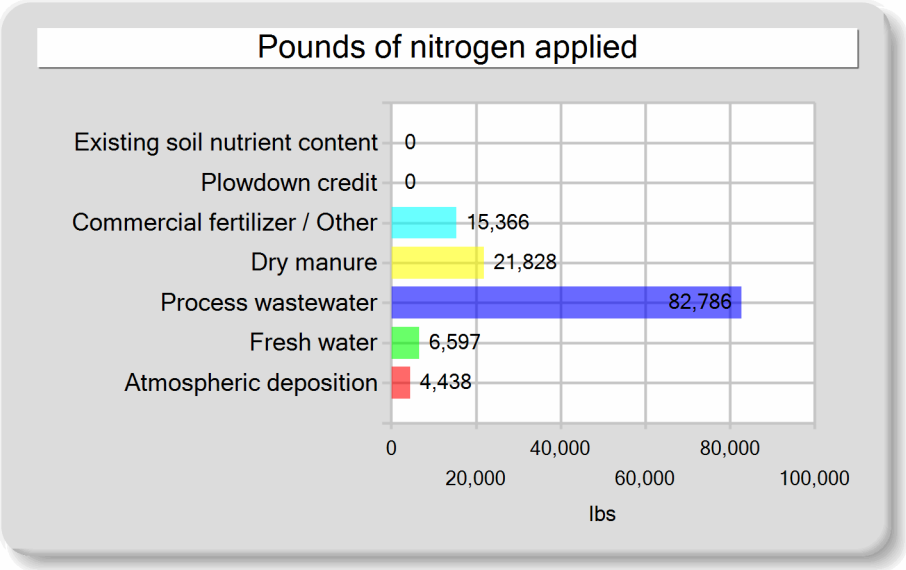
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

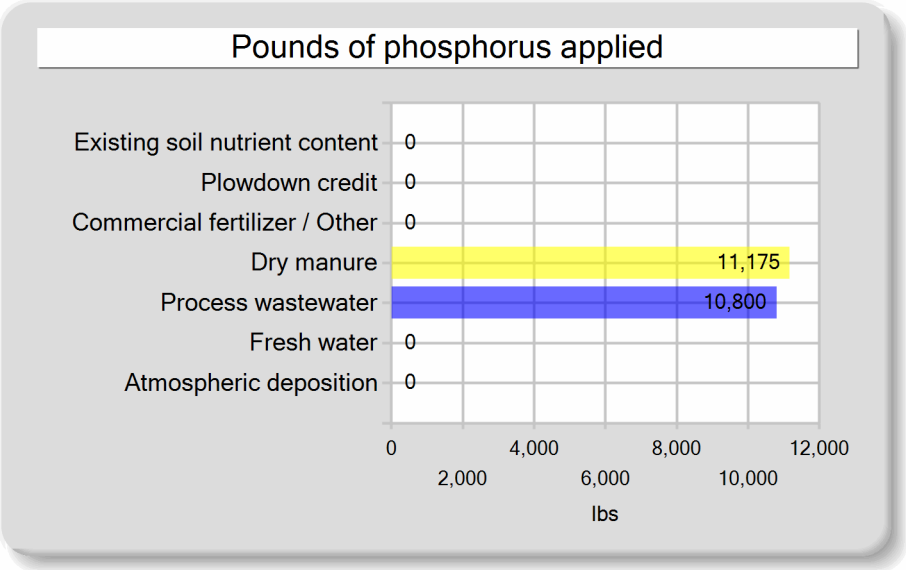
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	15,365.60	0.00	0.00	0.00
Dry manure	21,827.60	11,174.80	34,678.80	178,120.00
Process wastewater	82,786.24	10,800.23	121,293.27	703,171.06
Fresh water	6,597.29	0.00	0.00	448,333.34
Atmospheric deposition	4,438.00	0.00	0.00	0.00
Total nutrients applied	131,014.73	21,975.03	155,972.07	1,329,624.40
Anticipated crop nutrient removal	98,908.00	17,230.80	85,813.00	0.00
Actual crop nutrient removal	104,954.56	14,299.86	152,741.50	612,345.04
Nutrient balance	26,060.17	7,675.17	3,230.57	717,279.37
Applied to removed ratio	1.25	1.54	1.02	2.17

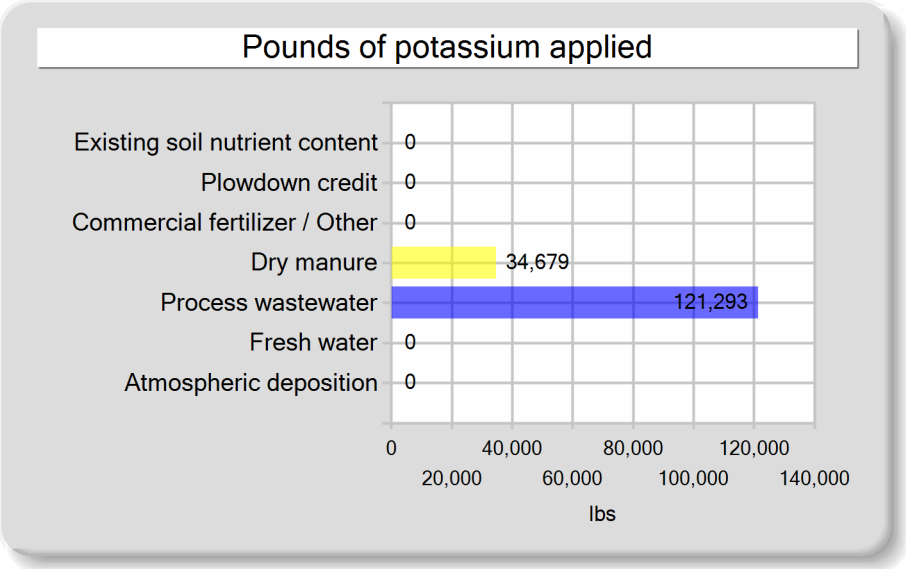
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

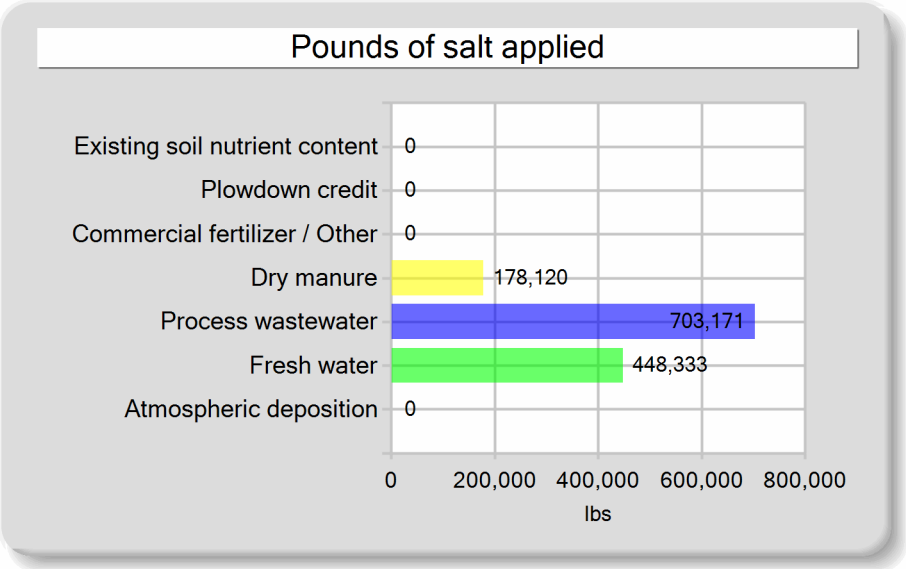


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

peter de Visser
peter de Visser (May 28, 2024 14:30 PDT)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Peter DeVisser	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
May 28, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE™
LABORATORY INC

P & A De Visser
21268 E Avena Rd
Escalon, CA 95320

Account# 00-0026267
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/03/2023 10:53

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23J1796-01	IW 1	Canal Water			10/25/2023 10:00
23J1796-02	IW 2	Canal Water			10/25/2023 10:00
23J1796-03	IW 3	Canal Water			10/25/2023 10:00
23J1796-04	IW Almonds	Canal Water			10/25/2023 10:00
23J1796-05	DW Milk Barn	Drinking Water			10/25/2023 10:00

Default Cooler Temperature on Receipt °C: 3.9

Containers Intact
COC/Labels Agree
Preservation Confirmed
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

P & A De Visser
21268 E Avena Rd
Escalon, CA 95320

Account# 00-0026267
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/03/2023 10:53

Sample Results

Sample: IW 1
23J1796-01 (Water)

Sampled: 10/25/2023 10:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.42**	mmhos/cm	0.01	1		10/31/23 16:44	SM 2510 B		BEJ1088
Electrical Conductivity umhos	**422**	umhos/cm	10.0	1		10/31/23 16:44	SM 2510 B		BEJ1088
Nitrate Nitrogen as NO3N	**7.9**	mg/L	0.1	1	10	10/27/23 01:52	EPA 300.0		BEJ1094
Temperature	**25.0**	units	0.0	1		10/31/23 16:44	SM 4500-H+	H	BEJ1088
pH	**8.3**	units	1.0	1		10/31/23 16:44	SM 4500-H+	H	BEJ1088
Total Filterable Solids (TDS)	**270**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:44	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**7.86**	mg/L	1.00	1		10/30/23 10:44	SM 4500-NH3 C		BEJ1110

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P & A De Visser
21268 E Avena Rd
Escalon, CA 95320

Account# 00-0026267
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/03/2023 10:53

Sample Results
(Continued)

Sample: IW 2
23J1796-02 (Water)

Sampled: 10/25/2023 10:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.71**	mmhos/cm	0.01	1		10/31/23 16:45	SM 2510 B		BEJ1088
Electrical Conductivity umhos	**712**	umhos/cm	10.0	1		10/31/23 16:45	SM 2510 B		BEJ1088
Nitrate Nitrogen as NO3N	**16.7**	mg/L	0.1	1	10	10/27/23 02:14	EPA 300.0		BEJ1094
Temperature	**25.0**	units	0.0	1		10/31/23 16:45	SM 4500-H+	H	BEJ1088
pH	**8.4**	units	1.0	1		10/31/23 16:45	SM 4500-H+	H	BEJ1088
Total Filterable Solids (TDS)	**460**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:46	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**16.7**	mg/L	1.00	1		10/30/23 10:46	SM 4500-NH3 C		BEJ1110

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P & A De Visser
21268 E Avena Rd
Escalon, CA 95320

Account# 00-0026267
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/03/2023 10:53

Sample Results
(Continued)

Sample: IW 3
23J1796-03 (Water)

Sampled: 10/25/2023 10:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.40**	mmhos/cm	0.01	1		10/31/23 16:46	SM 2510 B		BEJ1088
Electrical Conductivity umhos	**398**	umhos/cm	10.0	1		10/31/23 16:46	SM 2510 B		BEJ1088
Nitrate Nitrogen as NO3N	**4.7**	mg/L	0.1	1	10	10/27/23 02:36	EPA 300.0		BEJ1094
Temperature	**25.0**	units	0.0	1		10/31/23 16:46	SM 4500-H+	H	BEJ1088
pH	**8.3**	units	1.0	1		10/31/23 16:46	SM 4500-H+	H	BEJ1088
Total Filterable Solids (TDS)	**263**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:47	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**4.89**	mg/L	1.00	1		10/30/23 10:47	SM 4500-NH3 C		BEJ1110

The results in this report apply to the samples as received and were analyzed in accordance with the chain of custody document. This analytical report must be reproduced in its entirety. All results listed in this report are for the exclusive use of the submitting party. Dellavalle Laboratory, Inc. assumes no responsibility for report alteration, separation, detachment or third party interpretation.

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LABORATORY INC

P & A De Visser
21268 E Avena Rd
Escalon, CA 95320

Account# 00-0026267
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/03/2023 10:53

Sample Results
(Continued)

Sample: IW Almonds
23J1796-04 (Water)

Sampled: 10/25/2023 10:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.20**	mmhos/cm	0.01	1		10/31/23 16:48	SM 2510 B		BEJ1088
Electrical Conductivity umhos	**196**	umhos/cm	10.0	1		10/31/23 16:48	SM 2510 B		BEJ1088
Nitrate Nitrogen as NO3N	**1.3**	mg/L	0.1	1	10	10/27/23 05:31	EPA 300.0		BEJ1094
Temperature	**25.0**	units	0.0	1		10/31/23 16:48	SM 4500-H+	H	BEJ1088
pH	**8.0**	units	1.0	1		10/31/23 16:48	SM 4500-H+	H	BEJ1088
Total Filterable Solids (TDS)	**140**	mg/L	10.0	1		10/31/23 13:42	SM 2540 C		BEJ1105
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:48	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**1.34**	mg/L	1.00	1		10/30/23 10:48	SM 4500-NH3 C		BEJ1110

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P & A De Visser
21268 E Avena Rd
Escalon, CA 95320

Account# 00-0026267
Account Manager: Scott Sakamoto
Submitted By: Jeanette Sawyer

Received: 10/26/2023 16:45
Reported: 11/03/2023 10:53

Sample Results
(Continued)

Sample: DW Milk Barn
23J1796-05 (Water)

Sampled: 10/25/2023 10:00
Sampled By:

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.19**	mmhos/cm	0.01	1		11/01/23 11:08	SM 2510 B		BEJ1126
Electrical Conductivity umhos	**193**	umhos/cm	10.0	1		11/01/23 11:08	SM 2510 B		BEJ1126
Ammonia (as N)	*	mg/L	0.00	1		10/25/23 10:00	Field		BEJ1098
Nitrate Nitrogen as NO3N	**1.6**	mg/L	0.1	1	10	10/27/23 05:53	EPA 300.0		BEJ1094
Temperature	**25.0**	units	0.0	1		11/01/23 11:08	SM 4500-H+	H	BEJ1126
pH	**7.8**	units	1.0	1		11/01/23 11:08	SM 4500-H+	H	BEJ1126
Kjeldahl Nitrogen (TKN), Total	ND	mg/L	1.00	1		10/30/23 10:50	SM 4500-NH3 C		BEJ1110
Total Nitrogen	**1.82**	mg/L	1.00	1		10/30/23 10:50	SM 4500-NH3 C		BEJ1110

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10/26/23 16:45 23J1796

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

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Purchase Order No | Bill To: Acct # 2C26F Cons #

No. Samples: 5 No of Bottles: 10

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater [ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: P&A De Visser
Address: 21268 E Avena Rd
City: Escalon State: CA Zip: 95320
Telephone: 209 838-7383 Fax:
Cell/Email:
COPY TO: F&A Ag Services
& JSawyer@Fragservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Other:

Analysis and Bottles Required: (Please indicate Analysis)
DCWI TTV
DWWI TTV

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 1	10/25/23	started 10 AM	3.9	
2	2			2.9	
3	3			0.1	
4	Almonds			1.8	
5	DW Milk Barn			1.6	
6					
7					
8					
9					
10					

IR Thermometer SN: 200560723
Correction Factor: 0°C
Calibration Due: 12/22/2023
Location: Laboratory

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second	JLF	DLV	10-26-23 10:00	10-26-23 4:30
Third				
Fourth		DLV	10/26/6:45	

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:
Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature
Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

2023 Annual Report Peter De Visser

Final Audit Report 2024-05-28

Created:	2024-05-28
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAsg6G9jV-73e9QjBQJO7WBRglh1yMYxNz

"2023 Annual Report Peter De Visser" History

Document created by Vince Furtado (vfurtado@fragservices.com)
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